

CO₂ - AT₂ Short answer Test

course:- compiler design

course code:- C8A1402

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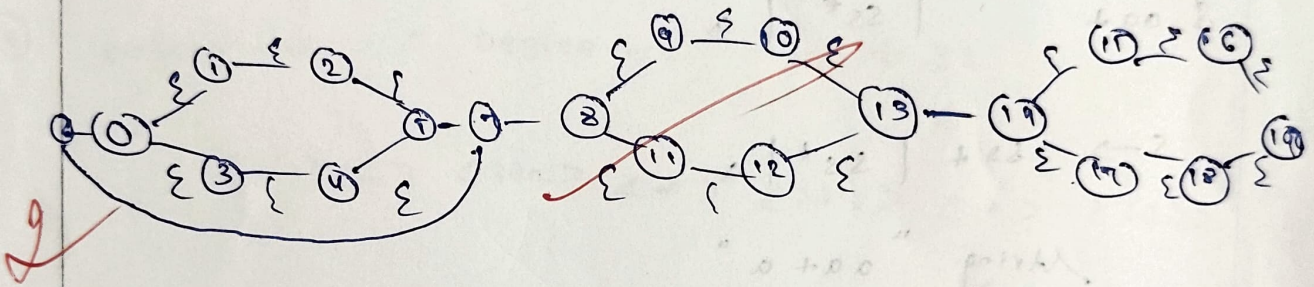
Reg no

Date:- 01/08/26

①

$a^1b^+ a (a^1b) (a^1b)$

$\frac{51}{100}$



②

$S \rightarrow (L) | a$

$S \rightarrow S' a$

$S \rightarrow (L) | \epsilon$

$L \rightarrow L' S | S$

$L \rightarrow S' L$

$L \rightarrow S' L | \epsilon$

$\text{first}(S) = \{ (, a \}$

$\text{first}(L) = \{ (, \epsilon \}$

$\text{follow}(S) = \{ , \}$

$\text{follow}(L) = \{ , \}$

③

$$S \rightarrow ss+ \mid ss^* \mid a$$

string

\$	ss+ ss* a	Shift
\$s	s+ ss* a	Shift
\$ss	+ ss* a	$s \rightarrow a$
\$aa	+ ss* a	Shift
\$aat	ss* a	

$$S \rightarrow ss+ \mid ss^* \mid a$$

String "aat+a"

$$S \rightarrow ss+$$

$$S \rightarrow ss^*$$

$$S \rightarrow a$$

$$\text{first}(S) = \{a\}$$

$$\text{first}(+) = \{+\}$$

$$S \rightarrow ss^*$$

$$s \rightarrow ss^*$$

$$s \rightarrow a$$

$$\text{first}(s) = \{a\}$$

$$\text{first}(+) = \{+\}$$

$$S \rightarrow a$$

$$\text{first}(S) = \{a\}$$

$$\text{follow}(S) = \{+, \epsilon\}$$

$$\text{follow}(s) = \{+, \epsilon\}$$

$$\text{follow}(+) = \{+, \epsilon\}$$

$$\text{follow}(a) = \{+, \epsilon\}$$

$$\text{follow}(+) = \{+, \epsilon\}$$

④

$$S \rightarrow sa \mid sb \mid c$$

$$S \rightarrow bs' \mid as' \mid \epsilon$$

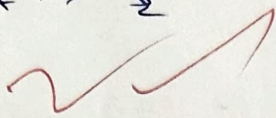
$$s \rightarrow as' \mid bs' \mid \epsilon$$

$$S \rightarrow \epsilon$$

$A \rightarrow abcD \mid abcE \mid F$

$A \rightarrow abcDA' \mid abcEA' \mid FA'$

$A \rightarrow \underline{\epsilon}$



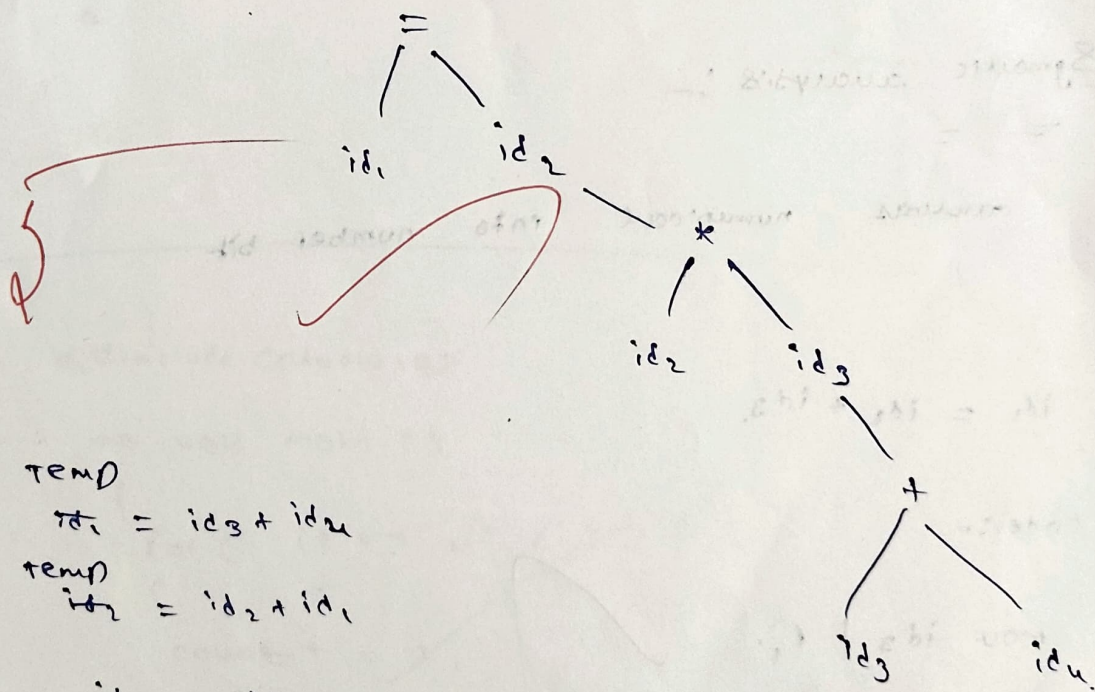
②

degr.

⑥

degree - F = Degree - c * 1.8 + 32

$\langle id_1 \rangle \langle = \rangle \langle id_2 \rangle \langle * \rangle \langle id_3 \rangle \langle + \rangle \langle id_4 \rangle$



temp

$id_1 = id_3 + id_4$

temp

$id_2 = id_2 * id_1$

$id_1 = id_2$

code :-

temp = id3 + id4

id1 = id2 * temp

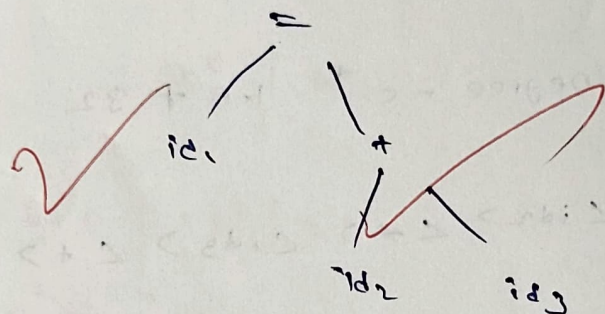
7

Sum = 0;

for (i = 1; i < 5; i++)

Sum = Sum + i

<id1> <id2> <id3> <id4> <id5>



Syntactic analysis :-

= =

convert numerical into number bit

8

id1 = id2 + id3.

code :-

mov id3 | r1

mov id2 | r2

Add r1 | r2

mov r2 | id1

$S \rightarrow aABb$, $A \rightarrow c | \epsilon$ $B \rightarrow d | \epsilon$

$\text{first}(S) = \{c\}$

$\text{first}(A) = \{c, \epsilon\}$

$\text{first}(B) = \{d, \epsilon\}$

follow
 $\text{first}(S) = \{ \}$

$\text{follow}(A) = \{c, \}$

$\text{follow}(B) = \{b, \}$

id

a

b

c

d

$S \rightarrow abcb$

$S \rightarrow c$

$A \rightarrow \epsilon$

$B \rightarrow d$

$B \rightarrow \epsilon$

S

A

B

9

`<#include <stdio.h>`

`int void main ()`

`{ for ( id = 1, * , * , ( )`

`{`

`count + = 1;`

`else.`

`count + = 0`

print (count, t=1, count t=0)

return 0;

}

}

10) left recursion
= = = =

the first the grammar to be

$A \rightarrow dAB$

$A \rightarrow BA'$

A

$A \rightarrow BA'$

- the non terminal for some of the production first non.

ex: $A \rightarrow B \mid B$

$F \rightarrow (E) \mid id.$

left recursion:

left factoring:

it should follow $N \rightarrow \alpha \beta$ in this statement.

- it act as left recursion.

(14)

 $S \rightarrow AB$ $S \rightarrow aA|a$ ~~abbc~~ abcc $S \rightarrow ca|c$

start

input

output

\$

abcc \$

Shift ($S \rightarrow AB$)

\$ AB

abcc \$

reduce $A \rightarrow a$

\$ aAB

abcc \$

Shift ($S \rightarrow aA|a$)

\$ sB|a

abcc \$

Shift
reduce ca,\$ ~~AB~~

abcc \$

reduce ~~B~~ $a \rightarrow c$

\$ B

abcc \$

reduce $a \rightarrow c$

\$ BC

abcc \$

Shift (B)

\$ C

cc

(15)

$S \rightarrow aAB$

$A \rightarrow bA|b$

$B \rightarrow c$

string - abbc

start	Input	output
\$	abbc \$	reduce ($S \rightarrow aAB$)
\$ A	abbc \$	
\$ AB	bbc \$	reduce ($A \rightarrow bbc$)
\$ AB	bbc \$	
\$ AB	bc \$	reduce ($A \rightarrow b$)
\$ AB	bc \$	reduce ($B \rightarrow c$)
\$ C	c \$	

$S' \rightarrow S$

$S \rightarrow aAB$

$A \rightarrow bA|c$

$B \rightarrow D$

$$S \rightarrow S$$

$$S \rightarrow aAB$$

$$aA \rightarrow bA \mid c$$

$$B \rightarrow d$$

$$(I_0, S) = I_1$$

$$S \rightarrow S$$

$$S \rightarrow aA$$

$$S \rightarrow bA \mid c \quad B \rightarrow D$$

$$\text{Given } (I_0, S) = I_2$$

$$S \rightarrow a, AB$$

$$\text{Given } (I_0, A) = I_3$$

$$A \rightarrow b, A$$

$$A \rightarrow C$$

$$\text{Given } (I_0, B) = I_4$$

$$B \rightarrow d$$

$$\text{Given } (I_1, S') = I_5$$

$$S \rightarrow A, B$$

$$\text{Given } (I_2, S) = I_6$$

$$A \rightarrow bA$$

$$A \rightarrow C$$

$$\text{Given } (I_1, B) = I_4$$

$$\text{Given } (I_1, B) = FG$$

$$\text{first}(S) = \{a\}$$

$$\text{first}(S) = \{a\}$$

$$\text{first}(A) = \{b, c\}$$

$$\text{first}(D) = \{d\}$$

$$\text{follow}(S) = \{ \$ \}$$

$$\text{follow}(S) = \{ \$ \}$$

$$\text{follow}(A) = \{ \$, B \}$$

$$\text{follow}(B) = \{ \$, d \}$$

$$S \rightarrow iEtSS' \mid a, s' \rightarrow es \mid \varepsilon, E \rightarrow b$$

$$\text{First } S = \{ b, a \}$$

$$\text{First } (S') = \{ e \}$$

$$\text{First } (E) = \{ b \}$$

$$\text{Follow } (S) = \{ \}$$

$$\text{Follow } (S') = \{ \}$$

$$\text{Follow } (E) = \{ \}$$

(12)

$$A \rightarrow aAB \mid aA \mid a$$

$$B \rightarrow bB \mid b$$

(a)

$$S \rightarrow ABC$$

$$A \rightarrow A \mid \varepsilon$$

$$B \rightarrow \pi \mid \varepsilon$$

$$C \rightarrow \beta \mid \varepsilon$$

$$\text{First } (S) = \{ a \}$$

$$\text{First } (A) = \{ \varepsilon, a \}$$

$$\text{First } (B) = \{ \pi, \varepsilon \}$$

$$\text{First } (C) = \{ b, \varepsilon \}$$

$$\text{Follow } (S) = \{ \beta \}$$

$$\text{Follow } (A) = \{ \pi, \beta \}$$

$$\text{Follow } (B) = \{ b, \beta \}$$

$$\text{Follow } (C) = \{ b, \beta \}$$

$$20) \quad S \rightarrow 1 S H \mid \epsilon A B C.$$

$$A \rightarrow a \mid b b D$$

$$B \rightarrow a \mid \epsilon$$

$$C \rightarrow b \mid \epsilon$$

$$D \rightarrow c \mid \epsilon$$

$$\text{First}(S) = \{1, a\}$$

$$\text{First}(A) = \{a, b\}$$

$$\text{First}(B) = \{a, \epsilon\}$$

$$\text{First}(C) = \{b, \epsilon\}$$

$$\text{First}(D) = \{c, \epsilon\}$$

$$\text{Follow}(S) = \{ \$ \}$$

$$\text{Follow}(A) = \{c, \$\}$$

$$\text{Follow}(B) = \{c, \$\}$$

$$\text{Follow}(C) = \{c, \$\}$$

$$\text{Follow}(D) = \{c, \$\}$$